

Code-Layout, Readability and Reusability

Date	2 July 2026
Team ID	XXXXX
Project Name	Rising Waters
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation is maintained throughout the Python code.
2	Proper File Structure	Files and folders are logically organized	Yes	Code is organized into separate modules for ML model and Flask application.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as Rainfall, WaterLevel, Humidity, and Flood clearly indicate their purpose
4	Function / Method Names	Functions are descriptively named	Yes	Functions like predict() and home() are descriptive and easy to understand.
5	Code Comments	Inline and block comments explain logic	Yes	Comments explain each major section of the program for better readability.
6	Modular Design	Code is split into reusable functions/modules	Yes	Machine learning logic, visualization, and web application are organized into separate sections.
7	No Redundant Code	Duplicate or unused code is removed	Yes	The code is concise with no unnecessary duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Basic functionality is implemented. Input validation and exception handling can be improved.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
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1	Dataset Preparation	Python (Pandas)	Data preprocessing	High
2	KNN Prediction Model	Scikit-learn	Flood prediction	High
3	K-Means Clustering	Scikit-learn	Flood risk analysis	High
4	Data Visualization	Matplotlib	Graph generation	High
5	Flask Web Application	Flask (Python)	User prediction interface	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized and easy to navigate
Readability	5	Clear naming conventions and comments improve understanding.
Reusability	5	ML model and Flask application can be reused in future projects.
Documentation / Comments	4	Sufficient comments are provided; additional documentation can further improve maintainability.
Overall Score	4.8	The project follows good coding practices with a clean structure, reusable modules, and readable code. Minor improvements can be made in error handling and documentation.